

AUTOMATION & MECHATRONICS

Intelligent **Parking** Management System

The Group 11

Zhejiang Gongshang University & University of Sussex

Sussex Artificial Intelligence Institute

SYSTEM FUNCTIONAL REQUIREMENTS



Capacity

Accommodation for at least 5 cars with real-time occupancy tracking.



Automated Gates

Servo-controlled barriers (GPIO 13/14) triggered by proximity and OCR.



Smart Billing

0.1 RMB/min rate with a 3-minute post-payment exit window.

HARDWARE MAPPING & GPIO

Peripheral Component	BCM Pin	Physical Pin	System Function
Front Entry Servo	GPIO 14	Pin 8	PWM Control: Entry Barrier Lift
Rear Exit Servo	GPIO 13	Pin 33	PWM Control: Exit Barrier Lift
IR Sensors (F/R)	GPIO 6 / 19	Pin 31 / 35	Vehicle Presence Detection
Warning LEDs (R/Y)	GPIO 21 / 20	Pin 40 / 38	Status Alerts (Full / Unpaid)

COMPUTER VISION ENGINE: ANPR PIPELINE

01 Spatial Localization (ROI Extraction)

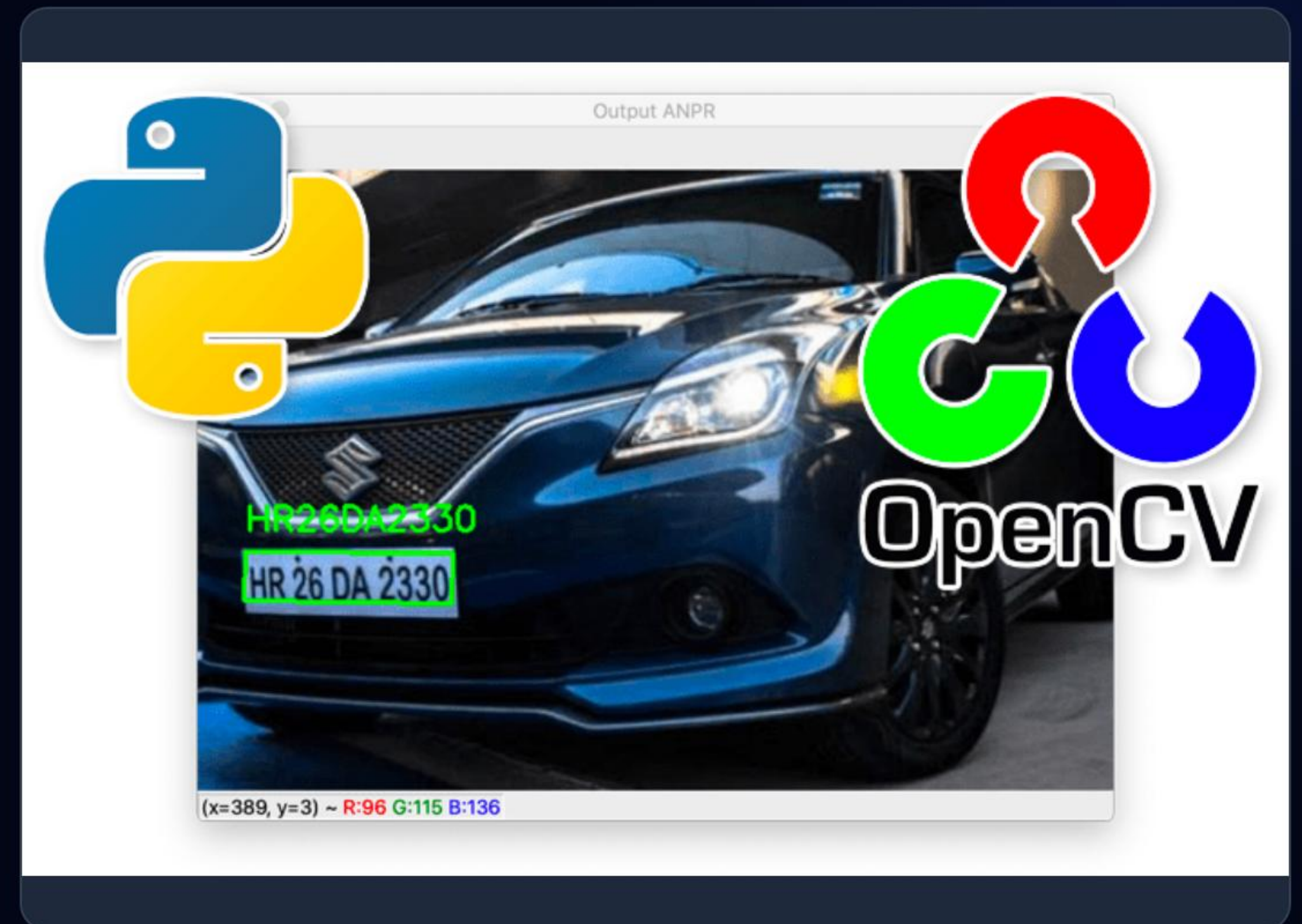
Cropping the central 50% of the video stream. Reduces computational overhead and eliminates non-relevant background artifacts.

02 Bilateral Edge-Preserving Smoothing

Replacing standard Gaussian blur with a bilateral filter to suppress surface noise while maintaining sharp character boundaries for OCR.

03 Adaptive Threshold Binarization

Overcoming uneven lighting and glare via local neighborhood pixel analysis, ensuring high-contrast input for the Tesseract engine.



MECHATRONIC CONTROL LOGIC



🔧 **Trigger Mechanism:** IR sensors (Active Low) trigger the camera scan logic via BCM 6/19.

🔒 **Safety Interlock:** Servo commands are sent via PWM duty cycle (10.0 for 90° lift) with non-blocking timing.

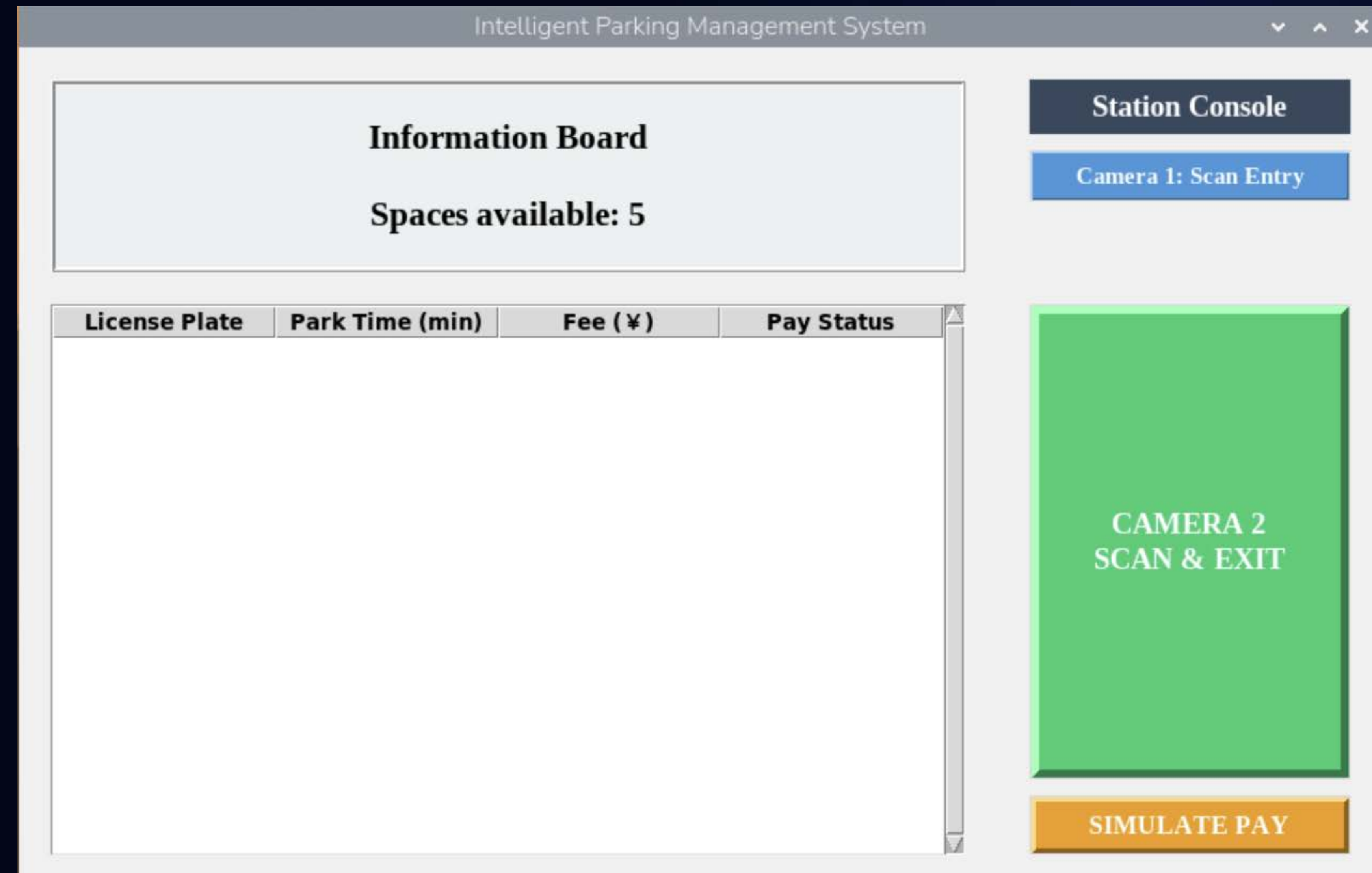
🕒 **Temporal Logic:** Exit gates only fire if `current_time - pay_time < 180s`.

💡 **Hardware Feedback:** GPIO 20 (Yellow) indicates pending payment status.

INTELLIGENT STATION CONSOLE

Developed with [Tkinter](#) for real-time monitoring and administrative override.

- **Live Data Stream:** Dynamic car tree view displaying license, time, and fee.
- **Simulation Controls:** Manual payment override for emergency scenarios.
- **Occupancy Display:** Information board turns **RED** automatically when spaces = 0.



SYSTEM EVALUATION & DEBUGGING

⚠ Engineering Challenge: Breadboard Continuity

During the integration phase, we identified a **discontinuity in the breadboard power rails**. Through systematic troubleshooting and multimeter verification, we re-routed the circuit to ensure stable 5V delivery to the dual servos and Raspberry Pi sensors.

This practical experience highlighted the importance of physical layer verification in mechatronic assembly.

Questions?

Thank you for your attention.

889HIZ Group Project Presentation

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